

Influence models of scholarly reference networks

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Abstract

1 Scholarly communications as reference networks

We model the social system of scholarly communications as a network of entities and their relationships (Crane, 2018). The entities comprise the sets:

$$\text{works: } \mathcal{W} = \{w_1, w_2, \dots, w_N\}, \quad (1)$$

$$\text{sources: } \mathcal{S} = \{s_1, s_2, \dots, s_M\}, \quad (2)$$

$$\text{authors: } \mathcal{A} = \{a_1, a_2, \dots, a_P\}, \quad (3)$$

$$\text{institutions: } \mathcal{I} = \{i_1, i_2, \dots, i_Q\}. \quad (4)$$

Entities have properties, e.g., works have a publication year $T(w_i)$, field and type, journals a publisher, authors an oeuvre, and institutions a country. Works have a reference list to earlier referenced works, corresponding to a list of tuples $(w_i, w_j, 1)$ that specify the edge list of a binary directed work-work network $(\mathcal{W} \times \mathcal{W})$ with adjacency matrix

$$\mathbf{C}^{\mathcal{W} \times \mathcal{W}} = c_{ij}^{\mathcal{W}\mathcal{W}}$$

The scholarly communication system forms a multipartite network where $\mathbf{C}^{\mathcal{W} \times \mathcal{W}}$ links other entity types through bipartite relations:

$$\text{publication of works } \mathcal{W} \times \mathcal{S} \quad (5)$$

$$\text{authorship of works } \mathcal{W} \times \mathcal{A} \quad (6)$$

$$\text{affiliation of authors on a work } \mathcal{A} \times \mathcal{I} \quad (7)$$

From the directed work-work citation network $\mathbf{C}^{\mathcal{W} \times \mathcal{W}}$, we project onto networks formed by the combinations of two or three entities:

$$\mathbf{C}^{\mathcal{S} \times \mathcal{S}} = \mathbf{B}^\top \mathbf{C}^{\mathcal{W} \times \mathcal{W}} \mathbf{B} \quad (8)$$

$$\mathbf{C}^{\mathcal{A} \times \mathcal{A}} = \mathbf{F}^\top \mathbf{C}^{\mathcal{W} \times \mathcal{W}} \mathbf{F} \quad (9)$$

$$\mathbf{C}^{\mathcal{I} \times \mathcal{I}} = (\mathbf{F}\mathbf{G})^\top \mathbf{C}^{\mathcal{W} \times \mathcal{W}} (\mathbf{F}\mathbf{G}) \quad (10)$$

$$\mathbf{C}^{\mathcal{S} \times \mathcal{A}} = \mathbf{B}^\top \mathbf{C}^{\mathcal{W} \times \mathcal{W}} \mathbf{F} \quad (11)$$

$$\mathbf{C}^{\mathcal{A} \times \mathcal{I}} = \mathbf{F}^\top \mathbf{C}^{\mathcal{W} \times \mathcal{W}} (\mathbf{F}\mathbf{G}) \quad (12)$$

$$\mathbf{C}^{\mathcal{S} \times \mathcal{I}} = \mathbf{B}^\top \mathbf{C}^{\mathcal{W} \times \mathcal{W}} (\mathbf{F}\mathbf{G}) \quad (13)$$

$$\mathbf{C}^{\mathcal{A} \times \mathcal{S} \times \mathcal{I}} = \mathbf{F}^\top \mathbf{C}^{\mathcal{W} \times \mathcal{W}} (\mathbf{F}\mathbf{G}) \quad (14)$$

where

$$\mathbf{B} \in \{0, 1\}^{N \times M}, \quad B_{ws} = \mathbf{1}[\text{work } w \text{ published in source } s] \quad (15)$$

$$\mathbf{F} \in \mathbb{R}^{N \times P}, \quad F_{wa} = \frac{\mathbf{1}[\text{author } a \text{ contributed to work } w]}{n_a(w)} \quad (16)$$

$$\mathbf{G} \in \mathbb{R}^{P \times Q}, \quad G_{ai} = \frac{\mathbf{1}[\text{author } a \text{ affiliated with institution } i]}{n_i(a)} \quad (17)$$

We summarise two approaches to iterative influence ranking over a journal citation matrix.

Let $\mathcal{J}$ be a set of n journals, and $\mathcal{P}^c$ and $\mathcal{P}^t$ two sets of articles published in $\mathcal{J}$. $\mathcal{P}^c$ is the census set published in the census-year window (Y_{start}^c, Y_{end}^c) and $\mathcal{P}^t$ the target set published in the target-year window (Y_{start}^t, Y_{end}^t) . Although $Y_{end}^c \geq Y_{end}^t$, $\mathcal{P}^c$ and $\mathcal{P}^t$ are not necessarily disjoint.

The census articles published in journal $\mathcal{J}_i$ have lists of references including some referring to target articles in journal $\mathcal{J}_j$. The citation matrix $\mathbf{C} \equiv c_{i,j}$ is an $n \times n$ matrix where element $c_{i,j}$ is the total count of references conferred by census-year articles in $\mathcal{J}_i$ to target-year articles in $\mathcal{J}_j$. Equivalently, $c_{i,j}$ counts citations received by $\mathcal{J}_j$ from $\mathcal{J}_i$ in the respective years. The citation matrix is fixed throughout the analysis, and can be viewed as a ‘snapshot’ or a ‘sample’ of the acknowledgements from census-year articles in $\mathcal{J}_i$ to influence by target-year articles in $\mathcal{J}_j$. works, $\mathcal{W} = \{w_1, w_2, \dots, w_N\}$, sources $\mathcal{S} = \{s_1, s_2, \dots, s_M\}$, authors $\mathcal{A} = \{a_1, a_2, \dots, a_P\}$ and institutions $\mathcal{I} = \{i_1, i_2, \dots, i_Q\}$. The row-sum vector is $\mathbf{s} = \mathbf{C}\mathbf{e}$; $s_j = \sum_k c_{j,k}$ where $\mathbf{e}$ is the n -element column vector of ones. Element s_j is the sum of census-window references from journal $\mathcal{J}_j$ to all target-window articles in $\mathcal{J}$. It is useful to define the diagonal matrix $\mathbf{\Lambda} = \text{diag}(s_1, \dots, s_n)$ and the total number of references $R = \sum_j s_j$.

Influence ranking aims to compute systematically a quantitative property of journals, the influence vector $\mathbf{w} \equiv w_i$ ($i = 1 \dots n$), such that a reference from a more influential journal receives a higher weight. Two different approaches have been used to motivate the idea of relative “influence” among journals:

Closed input-output (Pinski and Narin, 1976): The quantity $\sum_k c_{k,j}/s_j = \mathbf{e}^\top \mathbf{C} \mathbf{\Lambda}^{-1}$ is the ratio of references to target publications in $\mathcal{J}_j$ over references from census publications in $\mathcal{J}_j$. This is an estimate of the volume ratio of target-year ‘input’ to census-year ‘output’ for $\mathcal{J}_j$. Define $\gamma = \mathbf{C} \mathbf{\Lambda}^{-1}$. When the influence weight of every journal modifies the value of citations in all journals we have the reflexive

relation $\gamma^T \mathbf{w} = \mathbf{w}$. If this equation has a solution, they are a family of vectors of unspecified length. A solution of interest can be established by scaling $\mathbf{w}$ so that $\sum_k w_k s_k = \sum_k s_k = R$.

Probabilistic or Markov Chain (Geller, 1978; Bergstrom, 2007): Rather than estimate the ratio of input to output, we can estimate the probability that a publication in $\mathcal{J}_i$ will reference publications in $\mathcal{J}_j$, which is $\Lambda^{-1} \mathbf{C}$, a stochastic matrix. Define $\gamma^* = \Lambda \mathbf{C} \Lambda^{-1}$. When the influence weights of all journals in this model, say $\mathbf{w}^*$, modify the value of citations to all journals we have the reflexive relation $\gamma^{*T} \mathbf{w}^* = \mathbf{w}^*$. Because γ^* is a stochastic matrix we have $\sum_k w_k^* = 1$.

Before we solve these relations, note that following Geller (1978), the two influence weights are simply related, since

$$\Lambda \mathbf{W} = \Lambda \gamma^T \mathbf{W} = \Lambda \left[\Lambda^{-1} \gamma^{*T} \Lambda \right] \mathbf{W} = \gamma^{*T} \Lambda \mathbf{W}$$

This implies that $w_k^* = (s_k/R)w_k$, so $w_k^* < w_k$ consistently with the observation that w_k^* are probabilities and $w_k^*/w_k \sim s_k$ reflects a reduced significance of relative reference volume in the Pinski-Narin input-output formulation. The journal influence formulations of Pinski and Narin (1976), Geller (1978) and Bergstrom (2007) thus yield weights that are closely related for a given citation matrix.

The influence weights defined above represent the influence of an entire journal. Journals that publish more articles in a given time frame have a larger influence than those that publish fewer. The idea of influence developed here aims to focus on status and remove size, so per-article influence is obtained by re-scaling the influence of journal i by the relative journal size, $(|\mathcal{P}_i^c| + |\mathcal{P}_i^T|) / (|\mathcal{P}^c| + |\mathcal{P}^T|)$.

To find the influence vector for a given $\mathbf{C}$, Pinski and Narin (1976) used the power iteration method. Their journal set $\mathcal{J}$ covers a small and sufficiently coherent field (Physics) so that $\mathbf{C}$ is a *primitive matrix*: $\mathbf{C}$ is non-negative and $\mathbf{C}^n > 0$ for some positive exponent of primitivity n . $\mathbf{C}$ is thus also irreducible and has a single positive, greatest, real eigenvalue. For such a matrix the power iteration method initialises a vector $\mathbf{b}^{(k=0)} = 1/n$, say. It then computes $\mathbf{x}^k = \gamma^T \mathbf{b}^{k-1}$ and normalises $\mathbf{b}^k = \mathbf{x}^k / |\mathbf{x}^k|$. Then $\mathbf{b}^k$ converges to the dominant eigenvector.

If $\mathbf{C}$ is not primitive, the Katz and PageRank iterations are convenient extensions that yield the dominant eigenvalue albeit with the need to introduce an *a priori* parameter to regularise the matrix (Vigna, 2016; Franceschet, 2011).

2 Journal and Institution Influence

The article sets $\mathcal{P}^c$ and $\mathcal{P}^T$ associated with the journal set $\mathcal{J}$ involve authors and institutions listed in the article affiliation metadata.

3 Set-theoretic presentation

Let us define the fundamental sets underlying the citation network:

$$\begin{aligned}
\mathcal{W} &= \{w_1, w_2, \dots, w_N\} \quad (\text{set of works/publications}) \\
\mathcal{J} &= \{j_1, j_2, \dots, j_M\} \quad (\text{set of journals/venues}) \\
\mathcal{A} &= \{a_1, a_2, \dots, a_P\} \quad (\text{set of authors}) \\
\mathcal{I} &= \{i_1, i_2, \dots, i_Q\} \quad (\text{set of institutions}) \\
\mathcal{U} &= \mathcal{J} \cup \mathcal{A} \cup \mathcal{I} \quad (\text{unified entity set})
\end{aligned}$$

4 Structural Relations

Journal assignment: $J \subset \mathcal{W} \times \mathcal{J}$, $(w, j) \in J \iff$ work w published in journal j

Authorship: $\alpha \subset \mathcal{W} \times \mathcal{A}$, $(w, a) \in \alpha \iff$ author a contributed to work w

Affiliation: $\beta \subset \mathcal{A} \times \mathcal{I}$, $(a, i) \in \beta \iff$ author a affiliated with institution i

Citation: $R \subset \mathcal{W} \times \mathcal{W}$, $(w_i, w_j) \in R \iff w_i$ references w_j

Temporal Constraint:

$$(w_i, w_j) \in R \implies T(w_j) < T(w_i)$$

5 Fractional Counting Framework

5.1 Work-Level Citation Weights

The fundamental observable is the **citation edge list**: a set of tuples (w_i, w_j, ω) where $\omega = 1$ for whole counting. For fractional counting at the entity level, we define transformation functions.

5.2 Entity Connection Weights

For each work $w \in \mathcal{W}$, define its connected entities and their fractional weights:

Journal weight: $\omega_J(w, j) = \mathbf{1}[(w, j) \in J]$

Author weight: $\omega_A(w, a) = \frac{\mathbf{1}[(w, a) \in \alpha]}{n_a(w)}$, $n_a(w) = |\{a \in \mathcal{A} \mid (w, a) \in \alpha\}|$

Institution weight: $\omega_I(w, i) = \frac{\sum_{a \in \mathcal{A}} \mathbf{1}[(w, a) \in \alpha] \cdot \mathbf{1}[(a, i) \in \beta]}{n_a(w) \cdot n_i(a)}$, $n_i(a) = |\{i \in \mathcal{I} \mid (a, i) \in \beta\}|$

5.3 Total Entity Weight per Work

The total entity weight for work w is:

$$\Omega(w) = \sum_{j \in \mathcal{J}} \omega_J(w, j) + \sum_{a \in \mathcal{A}} \omega_A(w, a) + \sum_{i \in \mathcal{I}} \omega_I(w, i)$$

5.4 Normalized Membership Matrix

Define the normalized membership matrix $\mathbf{M} \in \mathbb{R}^{N \times |\mathcal{U}|}$:

$$M_{wu} = \begin{cases} \frac{\omega_J(w,u)}{\Omega(w)} & \text{if } u \in \mathcal{J} \\ \frac{\omega_A(w,u)}{\Omega(w)} & \text{if } u \in \mathcal{A} \\ \frac{\omega_I(w,u)}{\Omega(w)} & \text{if } u \in \mathcal{I} \end{cases}$$

6 Projected Citation Matrices

The entity-level citation matrices are obtained by **projecting** the work-level citation relation through the membership matrix.

6.1 Work-Level Citation Matrix

Let $\mathbf{A} \in \mathbb{R}^{N \times N}$ be the work-work citation matrix:

$$A_{ij} = \mathbb{I}[(w_i, w_j) \in R]$$

6.2 Entity-Level Projection

The unified entity citation matrix is obtained by projection:

$$\mathbf{C}^{\mathcal{U}} = \mathbf{M}^{\top} \mathbf{A} \mathbf{M}$$

Element-wise, this represents the sum of fractional citation weights:

$$C_{uv}^{\mathcal{U}} = \sum_{(w_i, w_j) \in R} M_{w_i u} \cdot M_{w_j v}$$

6.3 Specialized Projections

$$\text{Journal-Journal: } \mathbf{C}^{\mathcal{J}\mathcal{J}} = \mathbf{M}_{\mathcal{J}}^{\top} \mathbf{A} \mathbf{M}_{\mathcal{J}}$$

$$\text{Author-Author: } \mathbf{C}^{\mathcal{A}\mathcal{A}} = \mathbf{M}_{\mathcal{A}}^{\top} \mathbf{A} \mathbf{M}_{\mathcal{A}}$$

$$\text{Institution-Institution: } \mathbf{C}^{\mathcal{I}\mathcal{I}} = \mathbf{M}_{\mathcal{I}}^{\top} \mathbf{A} \mathbf{M}_{\mathcal{I}}$$

$$\text{Cross-type: } \mathbf{C}^{\mathcal{J}\mathcal{A}} = \mathbf{M}_{\mathcal{J}}^{\top} \mathbf{A} \mathbf{M}_{\mathcal{A}}$$

7 Edge List Representation

For computational implementation, the citation network can be represented as a **weighted edge list**:

```
referencing_work | referenced_work | weight
-----
w_i              | w_j              | 1.0
```

The entity-level citations are then computed by aggregation:

$$\text{EntityCitation}(u, v) = \sum_{(w_i, w_j) \in R} M_{w_i u} \cdot M_{w_j v}$$

8 Properties and Guarantees

- **Weight Preservation:** Each work-level citation contributes exactly 1.0 total weight:

$$\sum_{u, v \in \mathcal{U}} M_{w_i u} \cdot M_{w_j v} = 1 \quad \forall (w_i, w_j) \in R$$

- **Membership Normalization:** Each work distributes its membership weight completely:

$$\sum_{u \in \mathcal{U}} M_{w u} = 1 \quad \forall w \in \mathcal{W}$$

- **Cross-Type Compatibility:** The framework naturally handles citations between different entity types.

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